

Xylosome

A **motion-controlled line-scan camera**, built as an artistic instrument. A line-scan sensor has no frame — it captures one row of pixels at a time and the image only exists because something moves. Here that something is a servo-driven axis, and **the artist draws the movement**: a speed curve on a pendant, executed exactly as drawn.

Which means the operator is not setting an exposure. They are **controlling time, and the passage of time inside a single image**. A line-scan sensor builds a picture one row at a time, so every column of the finished frame is a different moment — the horizontal axis of the image *is* a timeline. The drawn speed curve decides how much time each part of the frame is given: slow through a passage and it dilates, accelerate and it compresses. A scan can span a second or twenty-four hours, and the same subject reads completely differently depending on where in the sweep the operator chose to spend time.

It runs **monochrome by default**. What it produces is governed by a wide parameter space — four capture modes, the drawn velocity profile, the arc, the span, line rate, TDI depth, gain, scan direction — and colour is **one optional add-on among those**, a four-pass filter sequence selectable per capture, not the machine's purpose.

Everything below lives on one aluminium cart: four computers, an EtherCAT motion chain, a 240 V servo running off a 120 V field outlet, and an imaging chain whose sync loop is the least obvious part of the machine.

Compiled 2026-09-05 Motion stack Beckhoff EtherCAT Default capture single-pass monochrome

01 CAPTURE MODES & THE PARAMETER SPACE

The instrument is not one behaviour with settings attached — it is **four distinct capture modes**, each with its own definition of what a scan *is*, sitting on top of a settings tree that reaches from the gearbox ratio to the camera's analog gain. All of it is reachable with **two buttons and a dial**.

CAPTURE MODES — WHAT THE AXIS DOES

SETTINGS — SIX BRANCHES



scan

velocity curve over an arc

drawn spline · arc start/end · aspect ratio
standard speed · 1-pass or 4-pass
above centreline accelerates, below decelerates

timed

constant crawl over a span

field of view arc · total span
seconds to 24 h, log-scaled on the dial

static

lines only — the camera never moves

the subject supplies all the motion

jog

position the axis — nothing is captured

setup and framing only

ADD-ON — COLOUR

a per-capture switch on the scan mode

1 pass = monochrome (default) · 4 passes = filtered colour
saved with the preset, so a recipe carries its own mode

presets

a whole capture recipe, recalled in one action

camera

line rate 3 500 – 38 314 Hz · TDI 16/32/48/64/80/96
gain -10 ... +10 dB · scan direction fwd/rev
live over the camera bus; link/bit-depth is read-only

axis / motion

gear ratio · standard scan speed · accel · homing
gear ratio 50 makes HMI degrees = output-flange degrees

calibration

homing offset · zero · touch

network

wifi · ip · controller link

display

brightness · blanking

system

firmware · platform · devices

NOT SETTABLE TODAY

lines per degree — structurally fixed at ~150.6
aspect bar and dial are coupled both ways
maximum speed — hardcoded 300 deg/s
curve endpoints — locked to the centreline
no straight segment can be drawn

Three of the four modes capture; one only positions. The distinction the modes draw is what the *axis* contributes — a drawn curve, a constant crawl, or nothing at all, leaving the subject to supply the motion. Colour attaches to a capture as a pass count, and travels with the preset. The red panel is what a designer cannot currently reach: three parameters that are structurally fixed rather than merely unexposed.

THE ONE THAT CONSTRAINS IMAGE GEOMETRY

Spatial sampling density is fixed at roughly 150.6 lines per degree and no control on the pendant can change it — the aspect bar and the dial hands are coupled in both directions, so the ratio falls out of the geometry rather than being chosen. This matters beyond the interface: the one scan on record that ever delivered its full planned line count ran at a *lower* density. If the chain cannot sustain 150.6, that coupling constant is the thing to change, and it is an interaction-design decision as much as a software one.

Where each parameter actually lives

A parameter's owner determines how fast it can change and who is allowed to change it. This is worth knowing before designing any screen around one.

PARAMETER	OWNED BY	REACHES THE HARDWARE VIA	PERSISTENCE
Drawn curve, arc, aspect	Pendant HMI	Sent to the motion daemon as a profile when the capture is triggered	Saved in presets
Span (timed mode)	Pendant HMI	Becomes a flat constant-velocity profile — arc divided by span	Saved in presets
Pass count (1 or 4)	Pendant HMI	Selects single-pass monochrome or the four-pass filter sequence	Saved in presets
Standard scan speed	Pendant › axis settings	Sets the curve's centreline height	Persisted on the Pi
Line rate, TDI stages, gain, scan direction	The camera itself	Pendant → capture agent → serial link to the camera	Volatile — reverts on camera power cycle unless explicitly written to its power-on set
Gear ratio, soft limits, slave positions	Motion controller config	Read at daemon startup	File on the controller
Line-trigger behaviour	Motion controller	Written to the pulse terminal over the bus at every startup	Does not survive power-off — reconfigured every run

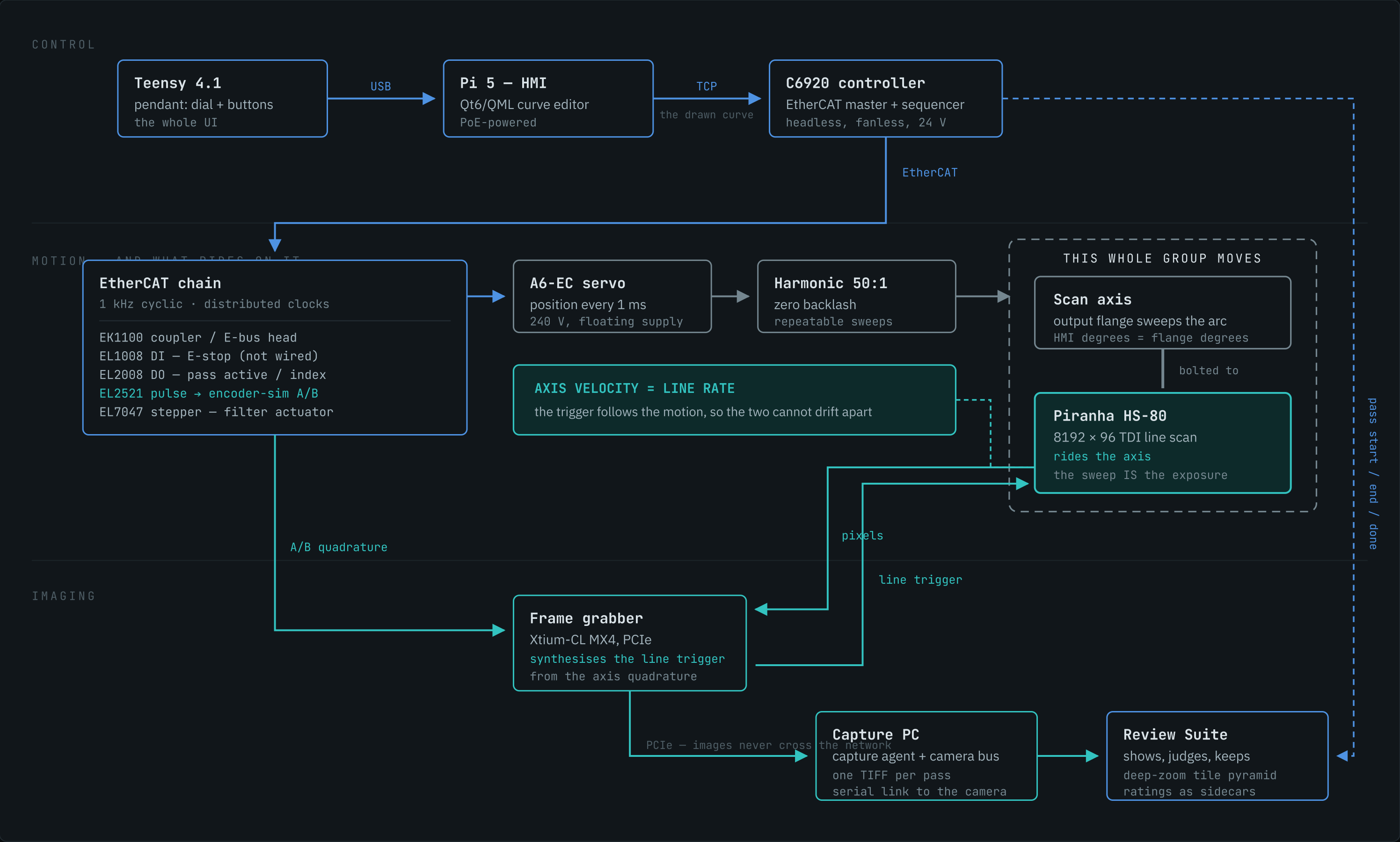
02 SIGNAL & MECHANICAL TOPOLOGY

Who talks to whom, and what physically moves. The single most important relationship on this page: **the camera is bolted to the scan axis, and the same axis motion that carries it also generates its line trigger**. Speed and sampling are not two settings — they are one mechanism seen from two ends. Power is drawn separately; overlaying both makes neither readable.

— control & network data

— imaging & line sync

— mechanical drive



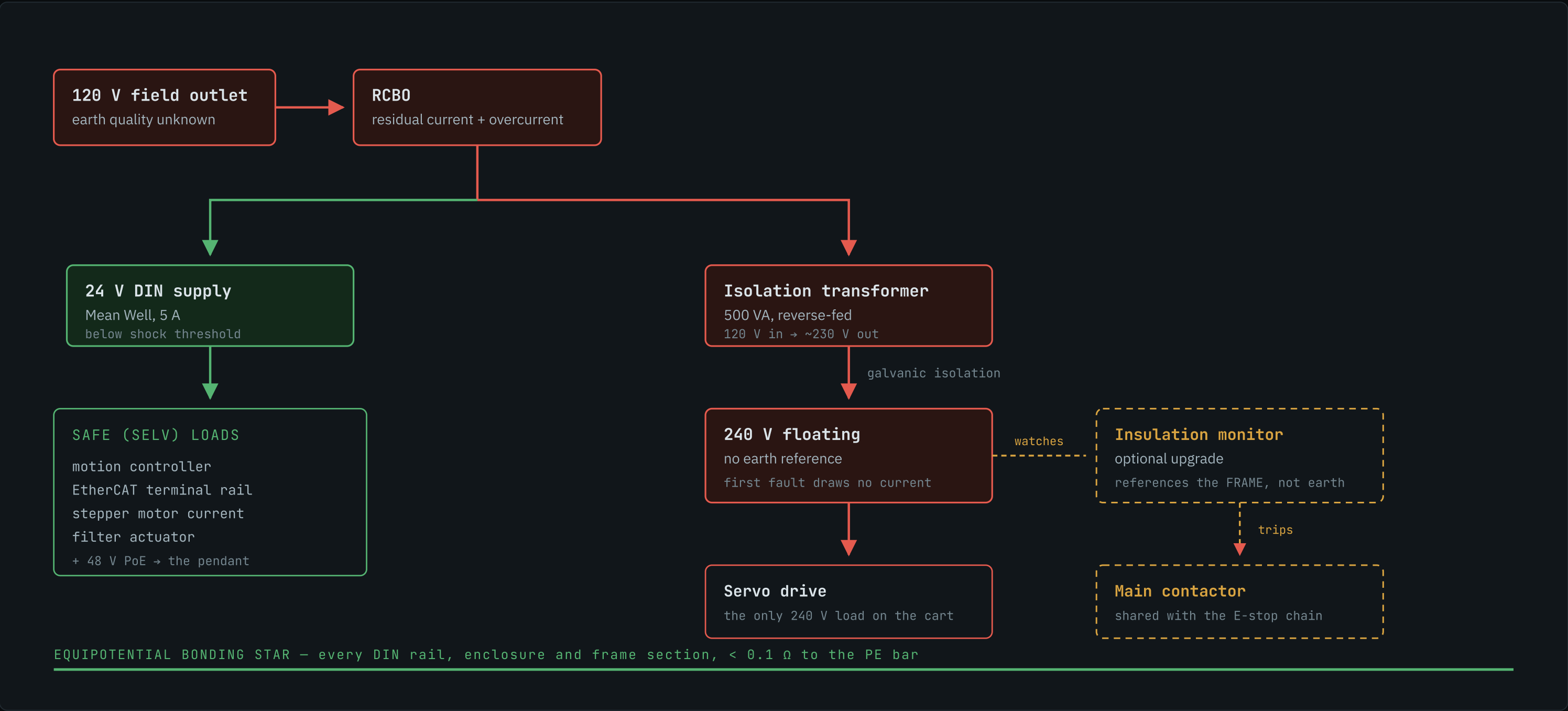
The camera rides the axis. It is bolted to the output flange, so the sweep is not pointing a camera at a subject — the sweep *is* the exposure, dragging the sensor across the scene one line at a time. And because the trigger is derived from that same axis motion, a faster sweep automatically clocks lines faster. The operator changes one thing and both follow.

The constraint that shapes everything: **the field gives you a 120 V outlet**, the servo needs 240 V, and the cart is an aluminium frame on rubber wheels that cannot rely on building earth.

mains – shock hazard

SELV – inherently safe

designed, not built



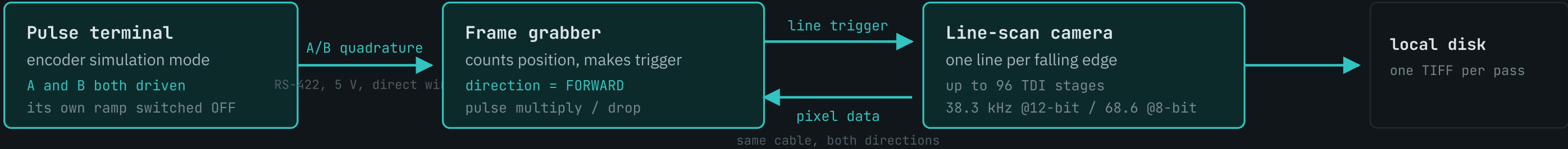
Two power worlds, deliberately separated. Everything except the servo runs at 24 V or 48 V — below the shock threshold — so the only real hazards on the cart are the 120 V inlet and the 240 V servo section. Dashed items are designed but not built.

THE TRAP THAT DEFEATS "JUST BOND THE FRAME"

Anodised aluminium is an insulator. The extrusion's surface is Al_2O_3 and T-slot joints are mechanical, not electrical — so the frame is *not* one continuous conductor, and different sections can sit at different potentials. Bonding has to be a deliberate star with anodize-piercing hardware from every rail and enclosure back to a central PE bar, then verified with a meter at under $0.1\ \Omega$ from every reachable spot. Anywhere reading open is an unbonded island. The frame is also open on all sides, so every mains terminal must be finger-safe or shrouded.

04 THE IMAGING SYNC LOOP

The least intuitive mechanism in the machine, and the one that produces the strangest failure when it is wrong. It is also where the coupling from section 02 is physically implemented: **position, not a clock, decides when each line is taken.** Sweep the axis twice as fast and twice as many lines are clocked per second, with no setting changed anywhere.



IF DIRECTION IS LOST, EVERY SCAN COMES BACK AS ITS OWN MIRROR

The axis returns over the same arc after every pass. If the grabber cannot resolve direction, it clocks that return as more forward lines. Two things must both hold: the terminal must drive B as well as A (in its default mode only A pulses, and direction is physically unknowable), and the grabber config must be set to FORWARD rather than its default of ignoring direction, which counts both ways.

Nothing wires from the I/O cable to the camera. The grabber is the pacemaker: it takes a position signal from the motion side, decides when each line should be clocked, and tells the camera over the control line. Position-locking the trigger this way is what keeps 96-stage TDI sharp — the line rate has to track velocity, or the image smears.

Compute & control

PART	WHAT IT DOES	WHERE & HOW	STATUS
Raspberry Pi 5 the pendant	The artist's control surface. Curve editor, mode picker, settings tree, presets. Records per-pass timing and the drawn curve, and exports one metadata file per session for post.	In the pendant box, PoE-powered over a single cable. Talks to the motion daemon over TCP, and falls back to a local simulation when no controller answers — so the interface stays usable on a desk.	RUNNING
Beckhoff C6920	Headless motion controller. Runs the EtherCAT master, the pass sequencer inside a 1 kHz cyclic loop, and a TCP server that broadcasts pass events.	Fanless, 24 V, in the panel. One NIC for the LAN, one dedicated to EtherCAT with no IP at all. Auto-starts on boot.	RUNNING
Capture PC	Owens the camera and the images. Runs the capture agent, the camera settings bus, a live-focus view, and the Review Suite. Writes one TIFF per pass.	Two grabber cards. One NIC carries all control traffic; a second is left free for a future fast image-offload path. The camera's serial link is single-occupant — one program at a time.	RUNNING
Review Suite	Shows, judges and keeps. Watches the capture folder, pairs files to passes by timing, builds a zoomable tile pyramid per TIFF, and stores ratings as sidecars. Originals stay read-only and there is no save step.	Desktop app; another client of the same daemon, so it learns pass boundaries for free.	INGEST BLOCKED ON WINDOWS
Teensy 4.1 pendant + Grayhill rotary encoder	The physical interface: one rotary encoder with push, plus hard buttons. The entire UI is reachable with two buttons and a dial — the constraint that shapes every screen.	On a custom carrier board, USB to the Pi. Not a network node. All board designs are constrained to parts the chosen fab can source.	RUNNING

Motion — the EtherCAT chain

One cable from the controller runs the whole chain at 1 kHz with clock synchronisation across every device.

PART	WHAT IT DOES	WHERE & HOW	STATUS
EK1100 coupler	Head of the terminal station, and the source of control power for everything behind it.	First on the DIN rail. Its power rail must be fed or the terminals report no control power — a failure that looks like a dead terminal.	FITTED
EL1008 — inputs	Eight digital inputs, intended for the emergency-stop chain and end limits.	On the rail, but switched off in configuration because the chain is not wired yet.	NOT WIRED
EL2008 — outputs	Eight digital outputs. Signals that a pass is active and which pass it is, so the rest of the cart can follow along.	On the rail, driven by the sequencer.	FITTED
EL2521 — pulse train	Emits the quadrature that paces the camera's line trigger. This is the single wire that couples geometry to sampling.	Differential 5 V straight into the grabber — no level shifting, no isolation. Reconfigured over the bus at every startup.	FITTED & RUNNING
EL7047 — stepper	Drives the filter actuator between passes, in four-pass mode only.	Needs two supplies: motor current on its own terminals, and control power through the coupler's rail.	AWAITING FILTER DESIGN
EL5101 — encoder in	Would tap the same quadrature feeding the grabber, giving an emitted-versus-delivered line count per pass.	Physically present and electrically compatible. Not yet mapped in configuration.	PRESENT, UNMAPPED
A6-EC servo drive	Drives the scan axis in cyclic synchronous position — the controller hands it a fresh target every millisecond, which is what makes a drawn curve reproduce faithfully.	Last on the chain. 240 V from the isolation transformer.	RUNNING

WORTH KNOWING BEFORE ANYONE REASONS ABOUT START-UP TIME

Getting the drive to accept clock synchronisation required a specific sequence performed while the bus is still in its pre-operational state, before the data map is built. **That mapping is legal only in pre-operational mode and does not survive power-off**, so it is redone on every single run rather than configured once. Cold start is therefore never instant, and that is structural rather than a bug.

Moving parts

PART	WHAT IT DOES	WHERE & HOW	STATUS
Harmonic drive, 50:1	Reduces the servo 50:1 with essentially zero backlash — which is what lets a drawn curve become a repeatable sweep, and lets four passes land on top of each other.	Between servo and output flange. With the ratio set in configuration, degrees on the pendant are degrees at the flange, so the artist's units are physical units.	FITTED
Scan axis	The sweep itself. Carries the subject or the optics through the arc while the camera clocks lines.	On the output flange. Returns over the same arc between passes — the reason direction sensing matters to the grabber.	RUNNING
Filter actuator NEMA 8 linear, tested to ~10 rev/s	Moves filters so each pass sees a different one. Only engaged in four-pass mode; idle for monochrome work.	Driven by the stepper terminal. The mechanism is being redesigned toward three filters behind the lens; rotary wheel versus linear slide is still an open decision.	REDESIGN IN PROGRESS
Multi-turn absolute encoder with battery backup	Would give a permanent absolute home with no switch and no hard stop — the axis knows where it is the moment it powers on.	Replaces the current wake-at-zero behaviour. Needs the battery cable fitted, the mode set, a first-connect fault cleared, and the offset taught once.	DECIDED, CABLE ORDERED
Aluminium extrusion cart	Carries everything. Open frame on rubber wheels, roughly a 30 × 24 inch panel area.	Open on all sides — no shielding, so EMC relies on physically separating the drive and motor region from signal wiring, and every mains terminal must be shrouded.	BUILT

Imaging

PART	WHAT IT DOES	WHERE & HOW	STATUS
Piranha HS-80 line-scan camera	8192 pixels wide, up to 96 TDI stages, 7 μm pixels, 8- or 12-bit. It has no shutter concept — it reads one line every time the trigger falls.	Control and image data share one cable. Settings arrive over a serial link and are volatile until deliberately written into the camera's power-on set.	RUNNING
Xtium-CL MX4 grabber	The actual pacemaker of the imaging chain. Consumes the motion-side quadrature, synthesises the line trigger, and streams pixels to disk.	PCIe card, I/O on a flying-lead cable. Its configuration file does not auto-load — today someone opens the vendor tool and loads it by hand at every cold start.	RUNNING
Capture agent	Names and files scans, serves a live-focus view with a focus metric, and exposes the camera's settings on a second bus so the pendant can reach them.	Two local service ports; also a client of the motion daemon, so it knows when a pass began.	RUNNING, NOT YET A SERVICE